Thapar Institute
of Engineering and Technology

CPG- 141

# CAPSTONE

# Semantic Drift in Open Vocabulary Object Detection Across Altitudes


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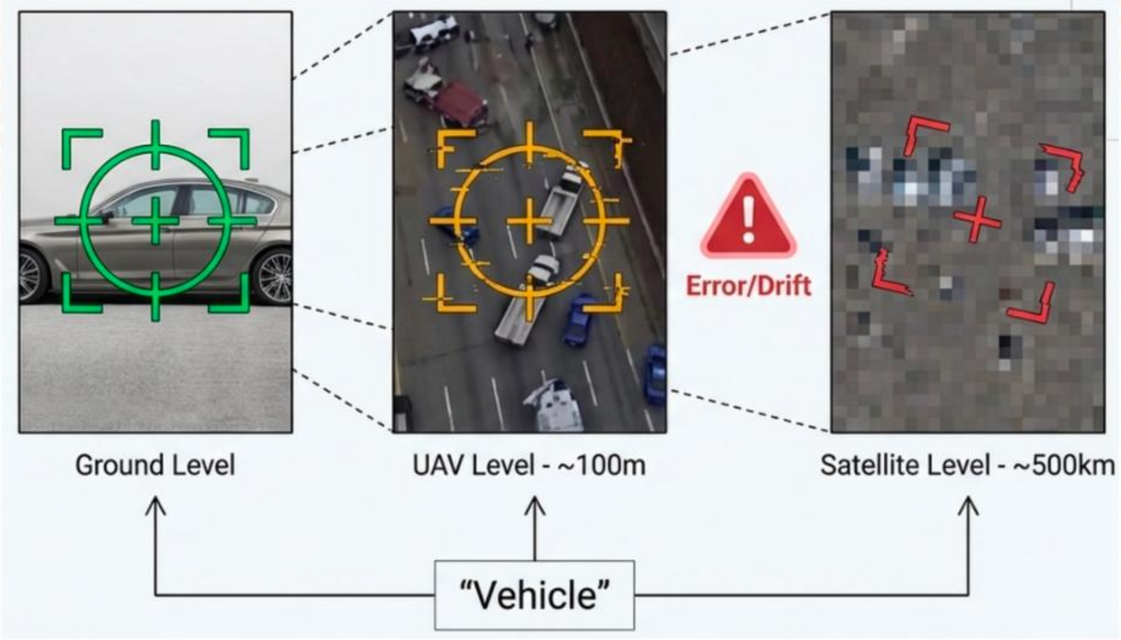
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# Project Overview

Open-Vocabulary Object Detection (OVOD) offers the revolutionary ability to detect novel objects using simple text prompts without retraining. However, this capability is fundamentally limited by the data foundation models are trained on.

## The Ground-Level Bias

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Foundation models are primarily trained on vast datasets of ground-level, natural-perspective imagery. This creates a strong bias towards objects as seen from a human eye-level, where visual cues are distinct and consistent.




## The Breakdown at Altitude

The Breakdown at Altitude
As altitude increases (UAV to Satellite), object appearance drastically changes due to scale, viewpoint, and atmospheric effects. The learned semantic relationship between the text prompt 'Vehicle' and its visual representation breaks down, causing significant detection failures known as 'semantic drift'.

# Need Analysis

**The Gap:** The lack of explicit modeling for altitude-dependent semantic variation.

## Cross-platform inconsistency

- Models fail to generalize between high-altitude (satellite) and low-altitude (UAV) imagery.
- Lack of "altitude-awareness" leads to critical misinterpretations.
- Example: Confusing "debris" with a "collapsed building" due to extreme shifts in scale, resolution, and perspective.

## Fragmented Domain Intelligence

- Lacks a "semantic bridge" to accurately fuse multi-source data in time-sensitive missions (e.g., Disaster Response, Defense).
- Produces inconsistent intelligence reports for the exact same target across different domains.
- Slows down coordination between large-scale monitoring and ground-level tactical response.

## Scalability & Automation Gap

- Current systems require expensive, manual retraining for every new sensor or flight altitude.
- Creates a major barrier to deploying truly autonomous, open-vocabulary monitoring systems.
- Hinders long-term, scalable deployment across dynamic environments like smart cities.

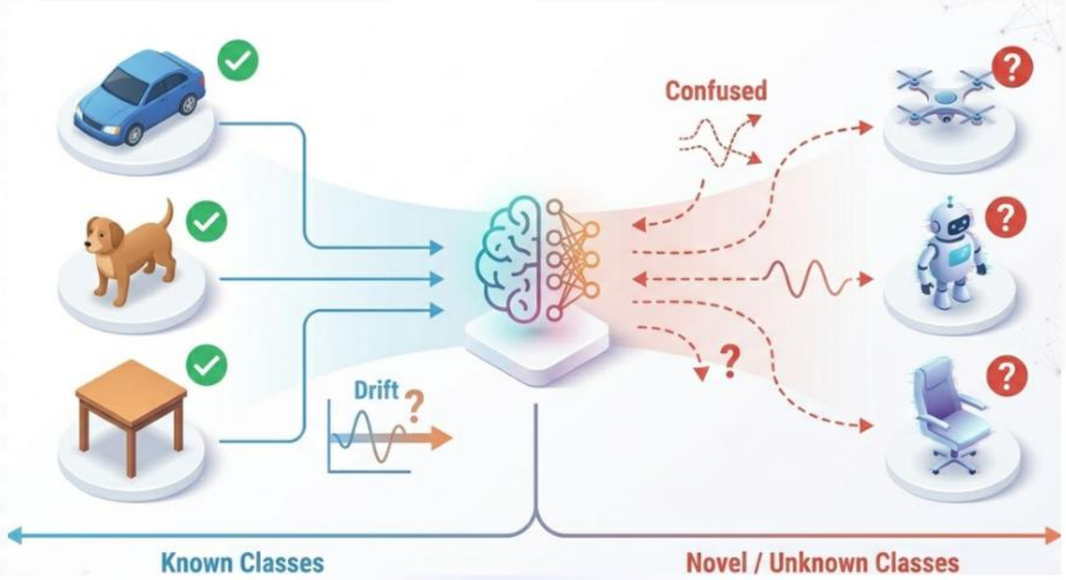
# LITERATURE SURVEY & RESEARCH GAP

| Research Area / Papers | Methodology / Approach | Key Assumptions | Limitations / Gaps Identified | Relevance to Our Work & Contribution |
|---|---|---|---|---|
| Open-Vocabulary Object Detection (OVOD) [1-4] (CLIP, RegionCLIP, GLIP, GroundingDINO) | Vision-Language Models (VLMs), Natural language descriptions for detection. | Semantic relationship between text and visual representation is consistent across domains. | Fails when visual context changes significantly; sensitive to scale, image quality. | Foundational models, but require adaptation for aerial domain's semantic drift. |
| Domain Adaptation in Object Detection [8-9] (Domain Adaptive Faster R-CNN, DA-DETR) | Aligns feature distributions between source and target domains. | Assumes invariant semantic meaning across domains. | Doesn't model how visual interpretation changes; struggles with fine-grained localization. | Addresses distribution shift, but insufficient for semantic meaning changes. |
| Aerial & Remote Sensing Domain [5-7] (xView, DOTA, VisDrone, UAVDT) | Datasets with significant variation in altitude, scale, viewpoint (Satellite vs. UAV). | Focus on data challenges, not explicit modeling of semantic changes. | Substantial performance degradation across domains due to scale, viewpoint, resolution. | Highlights the severe domain gap and need for specialized approaches. |
| Our Approach: Addressing Semantic Drift | Introduce 'Semantic Drift' to model changes in textual-visual relationship across altitudes. | Semantic grounding must be altitude-aware for robust detection. | (Addressed) Goes beyond feature alignment to ensure consistent semantic grounding. | **Our Contribution:** Formally defines and addresses the core problem of semantic drift in aerial OVOD. |

References: [1]Radfordetal.(2021)ICML.[2]Zhongetal.(2022)CVPR.[3]Lietal.(2022)CVPR.[4]Liuetal.(2023)arXiv.[5]Lametal.(2018)arXiv.[6]Zhuetal.(2019)ICCVW.[7]Zhuetal.(2021)TPAMI.
[8]Chenetal.(2018)CVPR.[9]Zhangetal.(2021)ICCV.[10]Fangetal.(2022)NeurIPS.[11]Torralba&Efros(2011)CVPR.[12]Sugiyamaetal.(2007)JMLR.

# PROBLEM STATEMENT

Existing Open-Vocabulary Object Detection (OVOD) models struggle with semantic drift when transitioning between satellite and UAV imagery. This drift caused by extreme altitude variations, diverse viewing angles, and domain-specific feature distortion leads to significant performance degradation, semantic confusion, and a lack of cross-domain robustness. Current detection frameworks fail to adequately quantify this inconsistency or provide altitude-aware adaptation strategies, resulting in unreliable object recognition for novel classes in remote sensing applications









# Objectives

- Formally define and quantify semantic drift in open-vocabulary object detection specifically across satellite and UAV imagery.
- Evaluate the impact of drift on detection performance, focusing on cross-domain generalization and semantic confusion.
- Design altitude-aware strategies, including embedding alignment, drift-aware loss functions, and prompt-based adaptation.
- Enhance cross-domain robustness and validate the approach using mAP, drift scores, and cross-altitude generalization metrics.

# ASSUMPTIONS

- Shared object concepts exist across altitude domains
- Same concept can have different visual grounding
- Vision-language embeddings can capture this shift
- Cross-domain performance drop is partly due to semantic drift
- A common ontology can be built

# CONSTRAINTS

- Direct comparisons are difficult due to varying dataset resolutions, camera angles, and object scales.
- Mismatched class names limit the shared ontology to a small set of reliable categories.
- Inconsistent bounding box annotations introduce noise that degrades accuracy and embedding quality.
- Extreme scale variations between altitudes make true semantic drift hard to isolate.
- Inherent baseline detector weaknesses with tiny objects may be conflated with semantic drift errors.

# Execution Plan

| Sr. No. | Activity | Month | July | | | | August | | | | September | | | | October | | | | November | | | | December | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Week no. | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 | 1 | 2 | 3 | 4 |
| 1 | Dataset Preparation and Ontology Mapping | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 2 | Baseline Open Vocabulary Detector | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 3 | Semantic Drift Measurement | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 4 | Drift Analysis & Failure Study | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 5 | Proposed Method-Alignment Module | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 6 | Evaluation & Comparison | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |
| 7 | Paper Writing & Submission | Plan | | | | | | | | | | | | | | | | | | | | | | | | |
| | | Actual | | | | | | | | | | | | | | | | | | | | | | | | |

*As of Aug 22, 2026*

# Methodology

## PHASE -1

- Select satellite and UAV imagery datasets to construct a shared, filtered vocabulary of 50–150 object categories.
- Standardize all image sizes and input formats across the different altitude regimes.

## PHASE -2

- Train three CLIP-based baseline models (satellite, aerial, and combined) that assign classes via embedding similarity.
- Evaluate these models to establish baseline in-domain and cross-altitude mean Average Precision (mAP) scores.

## PHASE -3

- Extract class embeddings from both domains to compute the mathematical semantic drift for each category.
- Rank the classes by their drift severity and visualize the embedding clusters using t-SNE or PCA.

## PHASE -4

- Analyze the correlation between the computed semantic drift and the observed drops in mAP performance.
- Conduct specific case studies to pinpoint failure modes like semantic confusion and cross-altitude misinterpretations.

## PHASE -5

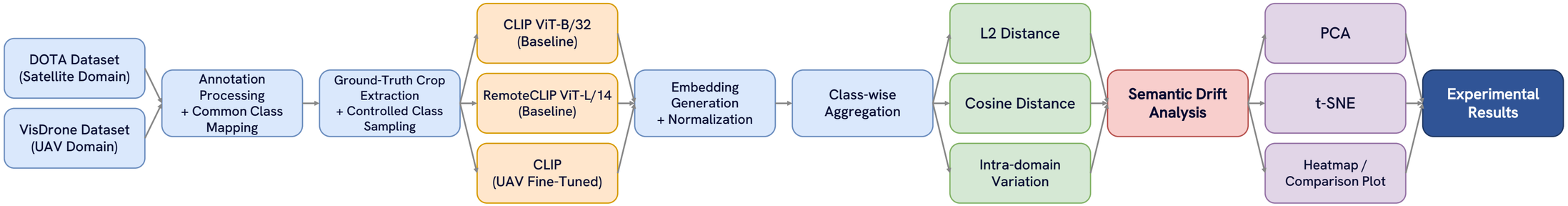
- Develop an alignment module that learns altitude-conditioned text embeddings based on the specific sensing domain.
- Process the original text embedding and an altitude indicator through a small MLP to generate context-aware representations.

## PHASE -6

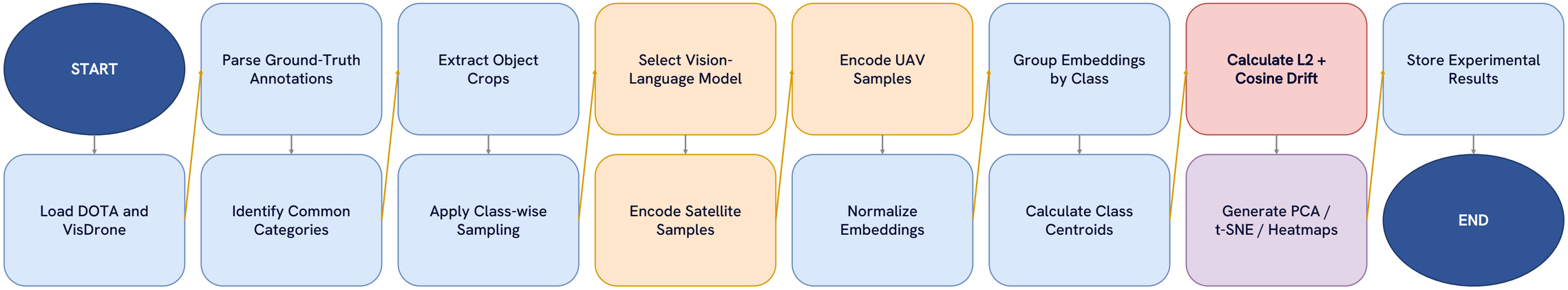
- Evaluate the alignment module by measuring overall mAP improvements.
- Quantify the method's success by tracking the reduction in semantic drift and analyzing per-class performance metrics.

# Architecture & Pipeline Design

## Overall Architecture — Semantic-Drift Analysis Framework (Fig. 4.1)




## Experimental Workflow — UML Activity Diagram (Fig. 4.2)



# Project Requirements




## HARDWARE

- High-VRAM Cloud GPUs (for training Swin-L/ViT backbones)
- Local machines for edge deployment testing



## SOFTWARE

- PyTorch & HuggingFace Transformers
- Python, OpenCV
- Standard CV evaluation libraries



## DATASETS

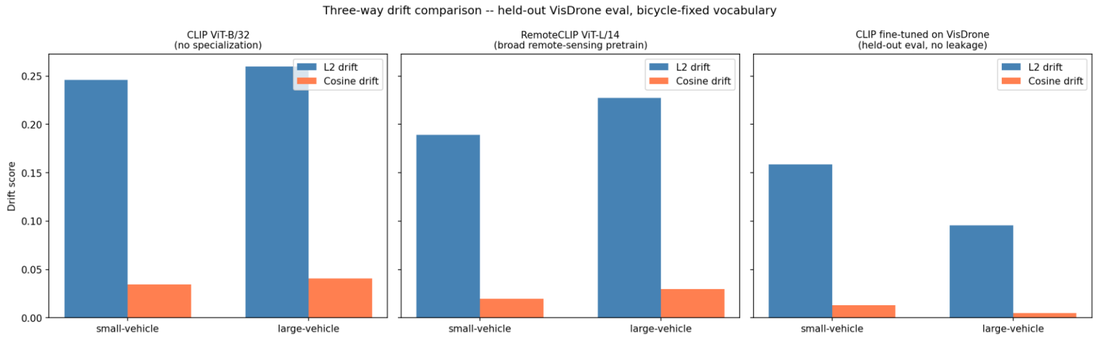
Satellite: DOTA, xView (streamed)
UAV: VisDrone, UAVid
Historical: Lightning AI Studio (Phase-1)

# Cost Analysis

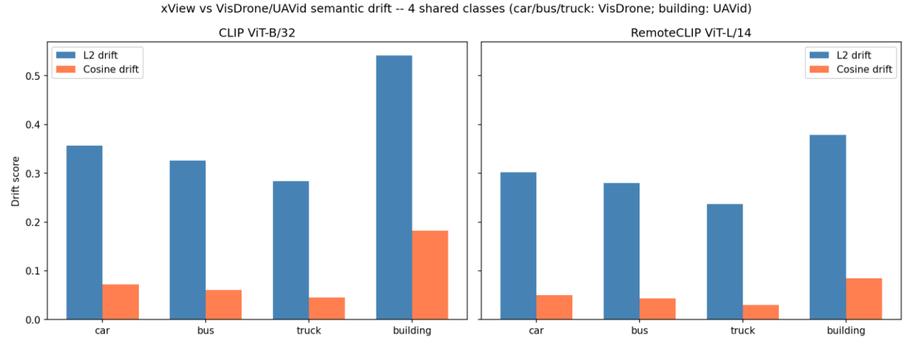
| Category | Details | Estimated Cost |
|---|---|---|
| **Compute** | Lightning AI Studio (Phase-1, cloud credits) + local RTX 3050 Laptop GPU (already-owned hardware) + allotted GPU | TBD |
| **Storage** | ~16 GB local disk for datasets, embeddings, checkpoints | $0 (existing hardware) |
| **Data acquisition** | DOTA, VisDrone, xView, UAVid — all open academic datasets | $0 |
| **Software & tools** | PyTorch, HuggingFace Transformers, open_clip, RemoteCLIP — all open-source | $0 |
| **Personnel** | 5-member student team | Not costed (academic capstone) |

***From the above breakdown it is clear that this project does not require any major expenses while being an impactful research contribution .***

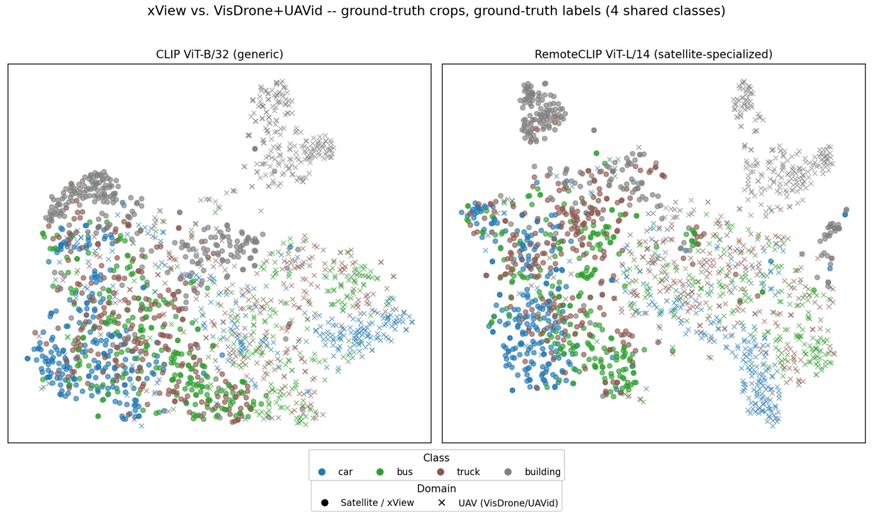
# Project Outcomes



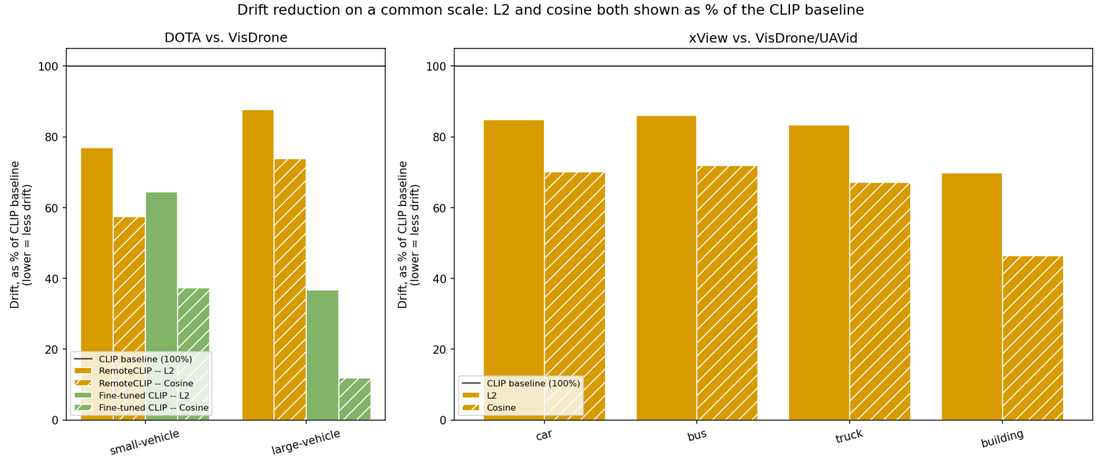

**DOTA ↔ VisDrone (3-way):** RemoteCLIP and our own fine-tuned CLIP both show consistently lower drift than general CLIP, on held-out data with no train/eval leakage.




**xView ↔ VisDrone/UAVid (4-class):** RemoteCLIP reduces drift on all 4 categories — building shows the largest drift and the largest reduction (30%) of anything measured.




**t-SNE, ground-truth crops:** the building class forms two visually distinct clusters by altitude — a direct geometric picture of its outsized drift score.



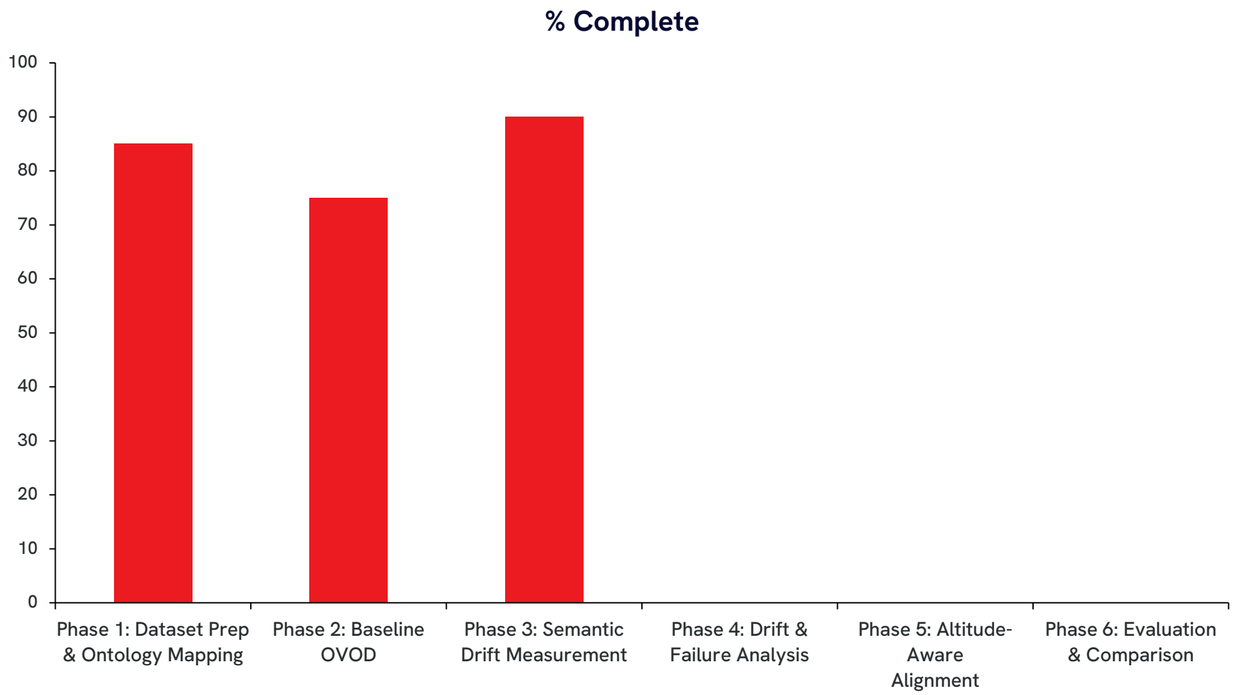

**Drift reduction, normalized:** L2 and cosine distance on one common scale (% of CLIP baseline) — specialization cuts drift 12–88% across every class and metric tested.

# Member Roles

| | |
|---|---|
| **Anoop Sachdev** | Experimentation & Evaluation |
| **Arnav Gupta** | Baseline & Model Implementation |
| **Kabir Oberoi** | Semantic Analysis & Visualization |
| **Khushpreet Kaur** | Data analysis & Preprocessing |
| **Siddhant Saxena** | Ablation Studies & Vocabulary formulation |

# Current Progress & Next Steps

## Completion by Approved Phase (Methodology, Sec. 1.8)




- 6 object categories tested
- 2 independent satellite datasets (DOTA, xView)
- 2 independent UAV datasets (VisDrone, UAVid)
- RemoteCLIP reduces drift on ALL 6 categories (12–30%)

## Next Steps

- Seed/sample-size robustness check
- Scale to full DOTA / VisDrone / xView
- Phase 4: drift-vs-mAP correlation study
- Phase 5: altitude-conditioned alignment module
- Phase 6: mAP + cross-altitude gap evaluation
- Integrate with teammate's SAM-based detection pipeline